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Inventor(s)	Sano; Takashi

Electronic device and wrist device

Abstract

An electronic device includes a ring-shaped antenna element, a conductor and a dielectric. The conductor has the same potential as GND. The dielectric is interposed between the antenna element and the conductor. The antenna element, the dielectric and the conductor are disposed so as to lie on top of one another in plan view.

Inventors:	Sano; Takashi (Fussa, JP)
Applicant:	CASIO COMPUTER CO., LTD. (Tokyo, JP)
Family ID:	77809069
Assignee:	CASIO COMPUTER CO., LTD. (Tokyo, JP)
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Primary Examiner: Leon; Edwin A.

Attorney, Agent or Firm: Scully, Scott, Murphy & Presser, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of U.S. patent application Ser. No. 17/205,142 filed on Mar. 18, 2021 which claims the benefit of priority from Japanese Patent Application No. 2020-053582 filed on Mar. 25, 2020 and Japanese Patent Application No. 2020-159198 filed on Sep. 24, 2020, the entire contents of which are incorporated herein by reference.

BACKGROUND

Technical Field

(1) The present disclosure relates to an electronic and a wrist device.

Description of Related Art

(2) A wristwatch capable of receiving satellite radio waves is disclosed in JP 2013-183437 A.

SUMMARY

(3) According to an aspect of the present disclosure, there is provided an electronic device including: a ring-shaped antenna element; a conductor having a potential same as GND; and a dielectric interposed between the antenna element and the conductor, wherein the antenna element, the dielectric and the conductor are disposed so as to lie on top of one another in plan view.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments, and together with the general description given above and the detailed description of the embodiments given below, serve to explain the principles of the present disclosure, wherein:

(2) FIG. 1 is a perspective view showing the main part of a timepiece body as an electronic device according to a first embodiment;

(3) FIG. 2 is a side sectional view of the timepiece body shown in FIG. 1;

(4) FIG. 3 is a perspective view showing the main part of a timepiece body according to a modification of the first embodiment;

(5) FIG. 4 is a perspective view of the timepiece body shown in FIG. 3 viewed from the side;

(6) FIG. 5 is a side sectional view of the timepiece body shown in FIG. 3;

(7) FIG. 6 is a sectional perspective view showing the main part of a timepiece body according to a modification of the first embodiment;

(8) FIG. 7 is a sectional perspective view showing the main part of a timepiece body as an electronic device according to a second embodiment;

(9) FIG. 8A is a sectional perspective view showing the main part of the timepiece body, wherein the inside diameter of a body case is larger than that shown in FIG. 7;

(10) FIG. 8B is a sectional perspective view showing the main part of the timepiece body, wherein the inside diameter of the body case is larger than that shown in FIG. 8A;

(11) FIG. 9 is a graph showing results of a simulation in which the inside diameter of the body case is changed, thereby showing tendencies of change in frequency and gain;

(12) FIG. 10A shows an example of a ring-shaped dielectric member that is substantially the same as an antenna element in shape;

(13) FIG. 10B shows an example of a dielectric member divided into a plurality of members;

(14) FIG. 10C shows an example of a dielectric member provided with recesses in the inner peripheral surface of the dielectric member;

(15) FIG. 10D shows an example of a dielectric member provided with the recesses shown in FIG.

10C, wherein other members are fitted in the recesses;

(16) FIG. **11** is a graph showing results of a simulation in which the width insulated by a dielectric member is changed, thereby showing tendencies of change in frequency and gain;

(17) FIG. **12** is a side sectional view showing an example in which an antenna element is fastened to a body case formed of a metal material;

(18) FIG. **13** is an enlarged view of a region XIII enclosed by a chain line in FIG. **12** and shows an example of an antenna element fastening structure;

(19) FIG. **14** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12**;

(20) FIG. **15** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12**;

(21) FIG. **16** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12**;

(22) FIG. **17** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12**;

(23) FIG. **18** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12**;

(24) FIG. **19** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12**;

(25) FIG. **20** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12**;

(26) FIG. **21** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12**; and

(27) FIG. **22** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12**.

DETAILED DESCRIPTION

First Embodiment

(28) With reference to FIG. **1** and FIG. **2**, a first embodiment of an electronic device according to the present disclosure and a wrist device to which this electronic device is applied will be described. In this embodiment, the body of a timepiece (hereinafter “timepiece body”) is the electronic device, and as an example, a timepiece (wristwatch) configured by a belt attached to the body so as to be worn on a wrist is the wrist device.

(29) Although various limitations technically preferable are put on the embodiments or the like described below, the scope of the present invention is not limited to the following embodiments, illustrated examples or the like.

(30) FIG. **1** is a perspective view showing the main part of a timepiece body as an electronic device of this embodiment. FIG. **2** is a side sectional view of the timepiece body shown in FIG. **1**.

(31) As shown in FIG. **1** and FIG. **2**, a timepiece body **100** of this embodiment includes a short columnar hollow case (“body case **1**” in this embodiment), which opens vertically in the thickness direction (opens on the front and back of the timepiece). The shape of the timepiece body **100** is not particularly limited.

(32) In this embodiment, the body case **1** is formed of a hard resin, such as ABS resin, for example.

(33) The material forming the body case **1** is not limited to resin, and other various materials, such as ceramic, are usable, for example. Alternatively, various composite materials in which carbon filler, glass fibers or the like is/are mixed with various resins may be used. The body case **1** formed of such a composite material can enhance the strength of the body case **1**.

(34) On the front surface side of the body case **1** (visible side or upper side of a timepiece), an optically transparent windshield member **2** (e.g. crystal) formed of transparent glass or the like is disposed so as to cover the opening.

(35) On the back surface side of the body case **1**, a back cover **3** that closes the opening is disposed.

The back cover **3** may be formed of the same material as that of the body case **1**, or may be formed of a different material. For example, when the body case **1** is formed of a resin, the back cover **3** may be formed of a metal.

(36) In this embodiment, the back cover **3** is formed of a conductor, such as a metal, and this metal back cover **3** functions as GND too. That is, the back cover **3** is GND disposed below an antenna element **4**, which will be described later.

(37) Over (above) the upper portion of the body case **1**, the antenna element **4** constituting an antenna (e.g. patch antenna) is disposed. That is, the body case **1** is interposed between the back cover **3** as GND and the antenna element **4**. The antenna element **4** of this embodiment is a ring-shaped bezel member, and is configured to be resonant with radio waves of a desired frequency. The shape of the antenna element **4** is not limited to the illustrated example.

(38) The material forming the antenna element **4** is not particularly limited. However, when conductivity of the forming material is low (when resistivity thereof is high), a sufficient antenna gain may not be obtained. Hence, as the material forming the antenna element **4**, a metal material having a certain level of conductivity or higher (i.e. a certain level of resistivity or lower), such as SUS (stainless steel) or titanium, is preferably used. In addition, because it is not preferable that the timepiece body **100** as an electronic device become too heavy, the antenna element **4** is preferably formed of a material as light as possible.

(39) When the antenna element **4** is disposed on the upper portion of the body case **1**, the material forming the body case **1** affects antenna characteristics.

(40) For example, in the case of the body case **1** formed of a resin material, when the antenna element **4** formed of a metal is disposed on the upper portion of the body case **1**, and the metal back cover **3** is GND, a resin is present between the antenna element **4** and the GND that faces the antenna element **4**.

(41) Resin is generally an insulator(s) and also a dielectric(s) having a permittivity. When the antenna element **4** is disposed on the upper portion of the body case **1** formed of a resin, according to the permittivity, thickness, shape, adhesion, arrangement and so forth of the resin material, the frequency, matching, circularly polarized wave axis ratio, circularly polarized wave frequency and gain as an antenna change.

(42) When the body case **1** is formed of only a resin material, the influence on the antenna characteristics can be dealt with to some extent by adjusting the matching or the arrangement. However, for example, when conductive fibers, carbon filler or the like are/is mixed with a resin, the resistance causes loss, and the gain as an antenna remarkably decreases.

(43) That is, when the antenna element **4** is disposed on the upper portion of the body case **1**, an electric field is generated in a form of, from the antenna element **4**, passing through the body case **1** so as to face GND (e.g. back cover **3**). In the case of the body case **1** formed of a composite material in which carbon filler or the like is mixed with a resin, it is inferred that an electric field passing through the body case **1** causes loss.

(44) In this embodiment, a conductive part (conductor) **5** is disposed at a position that is on the upper portion of the body case **1** and faces the antenna element **4**. That is, in this embodiment, the conductive part **5** is disposed below the antenna element **4** but on the upper portion of the body case **1** at least.

(45) Thus, the conductive part **5**, which is a metal sheet or thin film, disposed between the body case **1** and the antenna element **4** can prevent the generated electric field from passing through the portion where loss occurs. This can suppress occurrence of loss and maintain sensitivity of an antenna including the antenna element **4**.

(46) The conductive part **5** is preferably formed of a material having a lower resistance than the material of the body case **1** has. More specifically, the material forming the conductive part **5** is a metal, preferably a metal having a lowest possible resistance. To be more specific, aluminum or the like is suitably used. The material forming the conductive part **5** is not limited thereto, and various

materials are usable as long as they are low-resistance conductors.

(47) The thickness of the conductive part 5 is not particularly limited. The conductive part 5 may be formed of a sheet-shaped member, foil or the like that is pasted or a thin film formed by any type of vapor deposition, plating or the like.

(48) It is assumed that, as the material of the body case 1, a material selected from various materials according to the use or the like of an electronic device, such as the timepiece body 100, is used. It is not preferable that performance of the antenna element 4 as an antenna differ depending on the material of the body case 1.

(49) In this embodiment, the conductive part 5 is disposed at least on the upper portion of the body case 1, thereby being interposed between the body case 1 and the antenna element 4. This can prevent the body case 1, which is disposed under the conductive part 5, from affecting the antenna element 4, and enhance the degree of freedom in the material forming the body case 1.

(50) The conductive part 5 is disposed, of the body case 1, at least on the upper portion (upper end surface) where the body case 1 faces the antenna element 4, but in this embodiment, as shown in FIG. 2, the conductive part 5 is disposed so as to cover a region from the upper portion to the inner surface of the body case 1. With this configuration, the antenna element 4 can be connected to GND (e.g. back cover 3 that functions as GND), namely, can be grounded, through the conductive part 5.

(51) Thus, even when the antenna element 4 is disposed over the upper portion of the body case 1 formed of a resin, the antenna element 4 can be connected to GND through the connective part 5. Also, a dielectric member (dielectric) 6 disposed between the antenna element 4 and the conductive part 5 can contribute to appropriate wavelength shortening or the like, and cause the antenna element 4 to function as an antenna (e.g. patch antenna).

(52) The conductive part 5 may be configured to cover a region from the upper portion (upper end surface) through the inner surface to the bottom surface of the body case 1, the bottom surface being the inner surface of the back cover 3. In this case too, the antenna element 4 can be connected to the back cover 3, which functions as GND, through the conductive part 5.

(53) If the antenna element 4 is in direct contact with the conductive part 5, they short-circuit. Hence, an insulating member is disposed between the antenna element 4 and the conductive part 5. In this embodiment, the dielectric member 6 formed of a resin or the like is interposed between the conductive part 5 and the antenna element 4, thereby preventing the conductive part 5 and the antenna element 4 from contacting one another.

(54) It is preferable that the characteristics and shape of the dielectric member 6 be set according to a function(s) required for the antenna element 4. Thus, the antenna characteristics can be easily adjusted by adjusting the dielectric member 6.

(55) The material forming the dielectric member 6 is not particularly limited as long as it has insulating properties. For example, a resin having a high permittivity may be used.

(56) In a timepiece, it is important to make the inside of a case watertight. The dielectric member 6 may be configured to make the inside of the body case 1 watertight between the body case 1 and the antenna element 4.

(57) For example, when the conductive part 5 is a thin film formed by vapor deposition, plating or the like, the dielectric member 6 may be disposed on the conductive part 5, and pressed by the antenna element 4 as a bezel from the above.

(58) The conductive part 5, which is disposed on the upper surface of the body case 1, may not be disposed to the outer peripheral end thereof, and the dielectric member 6 may be configured to make the inside of the body case 1 watertight at a portion where the conductive part 5 is not disposed.

(59) The dielectric member 6 has a width that can stably support the antenna element 4 such that the antenna element 4 does not contact the conductive part 5, and does not necessarily cover the entire area of the conductive part 5 disposed on the upper portion of the body case 1.

(60) When the dielectric member **6** is not required to have a function of keeping the water-tightness of the body case **1**, the dielectric member **6** may not be disposed over the whole circumference of the upper surface of the body case **1**.

(61) In the body case **1**, a substrate **7** provided with an electric circuit **71** is housed. The electric circuit **71** includes, for example, a resonant circuit, such as an LC circuit (not shown), and/or a detector IC mounted on the substrate **7**.

(62) More specifically, the substrate **7** is, for example, a resin substrate formed of a resin, and the electric circuit **71** is provided by forming a circuit pattern on the substrate **7**.

(63) The antenna element **4** is connected to the electric circuit **71** on the substrate **7** by a first connector **73**. The first connector **73** is, for example, a coil spring.

(64) The first connector **73** is for power supply, and the antenna element **4** is connected, through the first connector **73**, to an LC circuit or the like as the electric circuit **71** mounted on the substrate **7**.

(65) To the electric circuit **71** on the substrate **7**, a second connector **74** is connected from the antenna element **4**. On the lower surface of the substrate **7**, a connector member(s) **75** for electrically connecting the substrate **7** to the back cover **3** as GND, thereby grounding the antenna element **4**, is disposed.

(66) When, like this embodiment, the conductive part **5** is disposed on the inner surface of the body case **1** too, the antenna element **4** can be grounded without the connector member **75**, but the connector member **75** may be disposed at a predetermined position so that the antenna element **4** can be grounded more stably and certainly.

(67) Although not shown, other than the substrate **7**, various members are housed in the body case **1**.

(68) Examples of the members that are housed in the body case **1** include a display unit constituting a display (not shown) of the timepiece body **100** and a module (not shown) that operates each part of the timepiece body **100**.

(69) Next, operation and effects of the timepiece body **100** as an electronic device will be described.

(70) To form the timepiece body **100** of this embodiment, first, from the upper end surface to the inner surface of the body case **1**, a conductive foil or sheet is disposed or a thin film is formed, so that the conductive part **5** is provided. Then, the metal back cover **3**, which functions as GND, is attached to the opening part on the back surface side of the body case **1** so as to cover the opening. The back cover **3** as GND is electrically connected to the conductive part **5**.

(71) In the body case **1**, the substrate **7** equipped with the electric circuit **71** and so forth are housed. Between the substrate **7** and the back cover **3**, the connector member **75** is disposed so as to electrically connect the substrate **7** and the back cover **3**.

(72) The antenna element **4** as a metal bezel or the like is disposed above the conductive part **5** on the upper portion of the body case **1**. To the antenna element **4**, the first connector **73** for power supply and the second connector **74** for GND are connected, so that the antenna element **4** is connected to the substrate **7** through the first connector **73** and the second connector **74**. Thus, the antenna element **4** is electrically connected to the back cover **3** as GND.

(73) Then, the dielectric member **6** is disposed between the antenna element **4** and the conductive part **5** such that the antenna element **4** and the conductive part **5** do not contact one another.

(74) Further, the windshield member **2** is attached to the opening part on the front surface side of the body case **1** so as to cover the opening.

(75) At the time when power is supplied to the antenna element **4**, and the antenna element **4** functions as an antenna, if there is no conductive part connected to GND, an electric field is generated in the form of, from the antenna element **4**, passing through the body case **1** so as to face GND (e.g. back cover **3**). When the body case **1** is interposed between the antenna element **4** and GND, an electric field passing through the body case **1** causes a large loss, depending on the material forming the body case **1**.

(76) In this embodiment, the conductive part **5** is disposed between the body case **1** and the antenna element **4**. This can prevent the generated electric field from passing through the body case **1**, and suppress occurrence of loss.

(77) As described above, according to this embodiment, the timepiece body **100** as an electronic device includes: the ring-shaped antenna element **4**; the conductive part **5** formed of a conductor and having the same potential as GND; and the dielectric member **6** interposed between the antenna element **4** and the conductive part **5**, wherein the antenna element **4**, the dielectric member **6** and the conductive part **5** are disposed so as to lie on top of one another in plan view.

(78) As a specific configuration of this embodiment, the timepiece body **100** includes: the antenna element **4**; the body case **1** disposed below the antenna element **4**; the conductive part **5** disposed below the antenna element **4** but on the upper portion of the body case **1**; the dielectric member **6** interposed between the conductive part **5** and the antenna element **4**, thereby preventing the conductive part **5** and the antenna element **4** from contacting one another; and the back cover **3** electrically connected to the antenna element **4** and the conductive part **5** and functioning as GND.

(79) The characteristics of an antenna are affected by the materials, shapes, positions and so forth of members, such as a body case and GND, disposed around the antenna. That is, various peripheral members are formed of metals, resins, composite materials thereof or the like, the properties of these materials are various, such as being conductive, insulating, dielectric and magnetic, and the characteristics, permittivity, magnetic permeability, loss ($\tan \delta$) and conductivity of these materials and their anisotropies affect the antenna characteristics.

(80) Body cases **1** or the like are formable with various materials, which include various metals and resins. When body cases **1** are formed of different materials, their properties are different from one another.

(81) When an antenna element **4** is disposed on the upper portion or the like of a body case **1**, in order to make the antenna element **4** function as an antenna having desired characteristics, the antenna element **4** needs to be adjusted according to that body case **1** because the antenna characteristics are affected by the material forming the body case **1**. This raises a problem that general-purpose antennas cannot be used.

(82) For example, in JP 2013-183437 A, there is disclosed a wristwatch capable of receiving satellite radio waves. However, as described above, the characteristics of an antenna are affected by the materials, shapes, positions and so forth of members, such as a body case and GND, disposed around the antenna. Hence, it is still difficult to make an antenna function as an antenna having desired characteristics (frequency and gain).

(83) The configuration of this embodiment, however, makes it possible, even if the body case **1** is formed of a material that causes a large loss when an electric field emitted from the antenna element **4** passes through the body case **1**, such as a composite material in which carbon filler or the like is mixed with a resin, for the electric field not to pass through the body case **1**.

(84) This can suppress influence of the characteristics of the material of the body case **1**, and produce a high-gain antenna. That is, regardless of the material forming the body case **1**, occurrence of loss can be suppressed, and sensitivity of an antenna including the antenna element **4** can be maintained. This can enhance the degree of freedom in the material forming the body case **1** or the like, and let users enjoy rich variations in the external appearance and texture of the timepiece body **100**.

(85) In this embodiment, the conductive part **5** is a conductor film disposed on (at least a portion of) the upper surface and at least a portion of the inner surface of the body case **1**.

(86) When the conductive part **5** is constituted by a conductor film, the conductive part **5** can be easily configured by vapor deposition, plating or the like.

(87) When the conductive part **5** is disposed so as to cover the body case **1** to the inner surface thereof, the antenna element **4** can be connected to GND (e.g. back cover **3** that functions as GND) through the conductive part **5**.

(88) When, like this embodiment, the characteristics and shape of the dielectric member **6** are set according to the function required for the antenna element **4** as an antenna, desired antenna characteristics or the like can be obtained by only adjusting the dielectric member **6**, without, for example, changing the shape or the like of the body case **1**.

(89) The dielectric member **6** may double as a sealing member that makes the inside of the body case **1** watertight. That is, because the dielectric member **6** is a member formed of a resin or the like, the dielectric member **6** can be a sealing member by the resin being shaped into a waterproof packing or the like.

(90) This, without a separate member, the inside of the body case **1** can be made watertight by the member for preventing the antenna element **4** and the conductive part **5** from short-circuiting.

(91) When this timepiece body **100** is applied to a wrist device, such as a wristwatch, because the body case **1** are formable with various materials, the degree of freedom in design or the like is enhanced, and a wrist device rich in design can be produced.

(92) Regardless of the material forming the body case **1**, the antenna characteristics can be kept excellent.

(93) In this embodiment, the antenna element **4** is disposed over the upper portion of the body case **1**, but an exterior member that covers at least a portion of the antenna element **4** may also be disposed over the upper portion of the body case **1**.

(94) FIG. **3** to FIG. **5** show an example of the configuration of a timepiece body as an electronic device provided with an exterior member.

(95) FIG. **3** is a perspective view of a timepiece body having an exterior member as viewed from the upper surface side. FIG. **4** is a perspective view of the timepiece body shown in FIG. **3** as viewed from the lateral surface side. FIG. **5** is a side sectional view of the timepiece body shown in FIG. **3**.

(96) In FIG. **3** to FIG. **5**, the same members as those in FIG. **1** and FIG. **2** are denoted by the same reference signs, and descriptions thereof are omitted.

(97) As shown in FIG. **3**, a timepiece body **200** of this example includes a substantially-ring-shaped exterior member **8**. In this example shown in FIG. **3** or the like, the exterior member **8** covers the whole antenna element **4**.

(98) The exterior member **8** includes: a ring-shaped body (hereinafter “member body”) **81** disposed so as to cover the substantially-ring-shaped antenna element **4**; and a hanging part **82** hanging from the outer peripheral edge of the member body **81**.

(99) The exterior member **8** is formed of, for example, a resin or the like, and the timepiece body **200** having an excellent design can be produced by varying the color and/or texture.

(100) As shown in FIG. **3** to FIG. **5**, the exterior member **8** covers the antenna element **4** and the dielectric member **6** such that they are invisible from the outside.

(101) That is, as shown in FIG. **4** and FIG. **5**, the hanging part **82** of the exterior member **8** is disposed so as to cover the antenna element **4**, the conductive part **5** and the dielectric member **6** interposed between these. This can prevent the antenna element **4**, the conductive part **5** and the dielectric member **6** from being visible as viewed from the outer surface side, and produce a very beautiful external appearance.

(102) In the example shown in FIG. **4** and FIG. **5**, the hanging part **82** of the exterior member **8** is configured such that its outer peripheral surface is substantially flush with the outer peripheral surface of the body case **1**.

(103) More specifically, as shown in FIG. **5**, a step part **11** is formed by making the outer peripheral end of the upper end surface of the body case **1** one step lower than a portion closer to the center in a radial direction, and the hanging part **82** is housed in the step part **11**.

(104) With this configuration, the external appearance becomes excellent, and also because no step is generated between the body case **1** and the exterior member **8**, dirt or the like hardly accumulates.

(105) As shown in FIG. 5, in addition to the dielectric member 6, a waterproof O-ring 6a or the like that makes the inside of the body case 1 watertight may be provided. In this case, the conductive part 5 is disposed at most to the inner side of a position where the O-ring 6a is disposed.

(106) This can produce a more reliable watertight structure.

(107) In the example shown in FIG. 3 to FIG. 5, an antenna element 41 (metal bezel) includes a cylindrical part 411 and a flange 412 that projects outward from the lower end of the cylindrical part 411.

(108) The inside diameter of the cylindrical part 411 is approximately the same as that of the body case 1. In the cylindrical part 411, the windshield member 2 is fitted.

(109) How to attach the windshield member 2 is not particularly limited, but, for example, the windshield member 2 is press-fitted into the cylindrical part 411, thereby being fastened thereto. Between the windshield member 2 and the inner peripheral surface of the cylindrical part 411, a sealing member may be disposed.

(110) Fastening the windshield member 2 by press-fitting the antenna element 41 formed of a metal material into the cylindrical part 411 can achieve firm fastening.

(111) The flange 412 is disposed so as to face, of the body case 1, the upper portion (upper end surface) where the conductive part 5 is disposed. Between the lower surface of the flange 412 and the conductive part 5, the dielectric member 6 is disposed, thereby preventing the flange 412 of the antenna element 41 from contacting the conductive part 5.

(112) The member body 81 of the exterior member 8 is provided with through holes 83 at points along the circumferential direction, and into the through holes 83, screws 84 are inserted from above the exterior member 8 toward the body case 1.

(113) The exterior member 8 is screwed to the body case 1 (fastened thereto with the screws (84), which hereinafter may be referred to as “screw-fastened”) in a state of being insulated from the antenna element 41.

(114) That is, when the screws 84 formed of a metal material contact the antenna element 41, they short-circuit.

(115) To deal with this, for example, through holes (not shown) larger than the through holes 83 are formed in the antenna element 41, and ring-shaped members or the like constituted by insulators are fitted into the through holes, and then the exterior member 8 is screw-fastened. This can prevent the antenna element 41 from contacting the screws 84 even when the exterior member 8 is fastened with the screws 84 formed of a metal, and fasten members from the exterior member 8 to the body case 1 without causing short-circuiting.

(116) The exterior member 8 may be screw-fastened in the state of being insulated from the antenna element 41 so as to be in a conductive state with the conductive part 5. In this case, the conductive part 5 is provided with insertion holes having a size for the screws 84 to contact, and the screws 84 are inserted into the insertion holes of the conductive part 5. Thus, the screws 84 contact and are electrically connected to the conductive part 5.

(117) When the exterior member 8 is thus configured, the metal screws 84 may be GND.

(118) In this embodiment, the dielectric member 6 is provided as a single member, but not limited thereto.

(119) For example, like a timepiece body 300 shown in FIG. 6, a dielectric member 60 may be divided into two members of a first dielectric member 61 and a second dielectric member (second dielectric) 62. In the illustrated example, the first dielectric member 61 is disposed in contact with the upper portion of the body case 1, and the second dielectric member 62 is disposed in contact with the inner peripheral surface of the upper end portion of the body case 1.

(120) In the example shown in FIG. 6, a metal bezel that is an antenna element 42 includes a body 421 disposed so as to face the upper portion (upper end surface) of the body case 1, a first hanging part 422 hanging from the edge on the inner peripheral side of the body 421, and a second hanging part 423 hanging from the edge on the outer peripheral side of the body 421.

(121) In this case, the first dielectric member **61** is interposed between the conductive part **5** disposed on the upper portion of the body case **1** and the antenna element **42** (the body **421** of the antenna element **42**), thereby preventing the conductive part **5** and the antenna element **42** from contacting one another.

(122) The first dielectric member **61** may have a function as a sealing member that keeps the inside of the body case **1** watertight.

(123) The first dielectric member **61** has a width that can stably support the antenna element **42** such that the antenna element **42** does not contact the conductive part **5**, and does not necessarily cover the entire area of the conductive part **5** disposed on the upper portion of the body case **1**.

(124) When the first dielectric member **61** is not required to have a function of keeping the watertightness of the body case **1**, the first dielectric member **61** may not be disposed over the whole circumference of the upper surface of the body case **1**.

(125) The second dielectric member **62** is interposed between the conductive part **5** disposed on the inner surface of the body case **1** and the antenna element **42** (the first hanging part **422** of the antenna element **42**), thereby preventing the conductive part **5** and the antenna element **42** from contacting one another.

(126) The second dielectric member **62** may function as a member that adjusts the characteristics of the antenna element **42** as an antenna.

(127) In this case, desired antenna characteristics can be achieved by changing the size (cross sectional area, thickness, width, etc.), shape or the like of the second dielectric member **62**.

(128) In the example shown in FIG. **6**, the first hanging part **422** of the antenna element **42** is provided with slits **424** at predetermined intervals.

(129) The antenna element **42** has an inward flange **425** that projects inward in the radial direction, and on this inward flange **425**, the windshield member **2** is disposed.

(130) In this state, the antenna element **42** is press-fitted into, thereby being fastened to, the body case **1**. The slits **424** allow smooth press-fitting of the antenna element **42** into the body case **1**.

(131) The slits **424** can prevent the biting part of the antenna element **42** (bezel) on the body case **1** from operating as an extra antenna element.

(132) The dielectric member **6** may be provided with a first alignment part, and the antenna element **4** may be provided with a second alignment part at a position corresponding to the first alignment part in a state in which the antenna element **4** is attached to the dielectric member **6**, the second alignment part being fitted with the first alignment part. For example, a threaded portion **663** and a threaded portion **453** shown in FIG. **14** may correspond to the first alignment part and the second alignment part, respectively.

(133) This makes it possible to dispose the antenna element **4** on the dielectric member **6** without misalignment.

(134) The dielectric member **6** and the body case **1** may be formed in an integrated manner (as a single unit).

(135) In this embodiment, the back cover **3**, which is disposed below the antenna element **4** with the body case **1** in between, is formed of a conductor, such as a metal, and functions as GND, but GND is not limited to the back cover **3**.

(136) For example, when the substrate **7**, a dial plate of a display (not shown) or the like housed in the body case **1** is formed of a metal material or the like, the substrate **7**, the dial plate or the like may function as GND.

(137) When the timepiece body as an electronic device includes a solar panel (not shown), the solar panel may function as GND.

(138) The GND terminal of the electric circuit **71** in the body case **1** may be directly connected to the conductive part **5** disposed on the upper portion of the body case **1**.

(139) The position of GND is not particularly limited, and hence, depending on the member that functions as GND, GND may not be disposed below the antenna element **4**, and/or the body case **1**

may not be interposed between the antenna element **4** and GND. Such a case is included in the present disclosure.

Second Embodiment

(140) Next, with reference to FIG. 7 to FIG. 22, a second embodiment of an electronic device according to the present disclosure and a wrist device including this electronic device will be described. While in the first embodiment, the body case **1** is formed of a resin material or the like, in this embodiment, the body case is formed of a metal material, which is different from the first embodiment. Hereinafter, points different from those in the first embodiment will be described, in particular.

(141) In FIG. 7 to FIG. 22, the same members as those in the first embodiment are denoted by the same reference signs, and descriptions thereof are omitted.

(142) FIG. 7 is a sectional perspective view showing the main part of a timepiece body as an electronic device of this embodiment. In FIG. 7, among the members constituting the timepiece body, for example, the windshield member **2** is not shown.

(143) As shown in FIG. 7, as in the first embodiment, a timepiece body of this embodiment includes a short columnar hollow body case **10**, which opens vertically in the thickness direction (opens on the front and back of the timepiece). The timepiece body (the body case **10** of the timepiece body) is substantially ring-shaped in plan view from the above. The “ring-shaped” indicates not only a circular ring in plan view but also a wide range of various continuous shapes, such as an ellipse and a rectangle. The shape of the timepiece body (the body case **10** of the timepiece body) is not particularly limited.

(144) In this embodiment, the body case **10** is formed of, for example, a metal material, such as SUS or titanium. The material forming the body case **10** is not limited to those exemplified herein.

(145) In this embodiment, the body case **10** is formed of a conductor, such as a metal, and this metal body case **10** functions as a conductive part having the same potential as GND.

(146) On the front surface side of the body case **10** (visible side or upper side of a timepiece), the windshield member is disposed so as to cover the opening, and on the back surface side of the body case **10**, the back cover **3** that closes the opening is disposed. The back cover **3** may be formed of the same material as that of the body case **10**, or may be formed of a different material. For example, when the body case **10** is formed of a metal, the back cover **3** may be formed of a resin.

(147) When, like this embodiment, the body case **10** is formed of a metal material, the body case **10** itself is a conductive part. Hence, the conductive part of the first embodiment is unneeded, but may be provided additionally. In this case, the added conductive part is formed of a metal having a lower resistance than the metal forming the body case **10**. That is, for example, when the body case **10** is formed of SUS, titanium or the like, the added conductive part is formed of aluminum or the like. As compared with the case where the conductive part (i.e. a thin film of aluminum or the like) is not added, loss as an antenna can be reduced, and a more sensitive antenna can be produced.

(148) Over (above) the upper portion of the body case **10**, an antenna element **43** constituting an antenna (e.g. patch antenna) is disposed. As in the first embodiment, the antenna element **43** of this embodiment is a ring-shaped bezel member, and is configured to be resonant with radio waves of a desired frequency.

(149) The shape of the antenna element **43** is not limited to the illustrated example.

(150) The material forming the antenna element **43** or the like is the same as that in the first embodiment, and hence description thereof is omitted.

(151) The antenna element **43** is provided with a connector member **77** for connecting the antenna element **43** to the body case **10** as GND.

(152) When, like this embodiment, the body case **10** itself is a conductive part, the antenna element **43** can be grounded without the connector member **77**, but the connector member **77** is preferably disposed at a predetermined position so that the antenna element **43** can be grounded more stably and certainly.

(153) The connector member **77** is connected to the substrate **7** (circuit board) as GND through an electronic component **711** that is, for example, a 0Ω resistance (R), a coil (L) or a capacitor (C). The GND as a connection destination of the connector member **77** may not be the circuit board, but the circuit board is preferable because the antenna characteristics are likely to be exhibited by connecting the antenna element **43** to the circuit board through a coil (L) or a capacitor (C).

(154) The connection of the connector member **77** may be open.

(155) In this embodiment, the body case **10**, over the upper portion of which the antenna element **43** is disposed, is, as described above, a conductive part having the same potential as GND.

(156) Hence, when the antenna element **43** is disposed over the upper portion of the metal body case **10**, the antenna element **43** can be connected to GND through the body case **10**.

(157) However, if the antenna element **43** is in direct contact with the body case **10**, they short-circuit. Hence, an insulating member is disposed between the antenna element **43** and the body case **10**. In this embodiment, as in the first embodiment, a dielectric member **63** formed of a resin or the like is interposed between the body case **10** and the antenna element **43**, thereby preventing the body case **10** and the antenna element **43** from contacting one another.

(158) In this embodiment, the dielectric member **63** is ring-shaped so as to be substantially the same as the ring-shaped antenna element **43**, and as shown in FIG. 7, is preferable to lie on the antenna element **43**.

(159) Thus, in this embodiment, in the timepiece body including: the ring-shaped antenna element **43**; the body case **10** that functions as a conductive part having the same potential as GND; and the dielectric member **63** interposed between the antenna element **43** and the body case **10**, the antenna element **43**, the dielectric member **63** and the body case **10** lie on top of one another in plan view.

(160) The dielectric member **63** has a width that can stably support the antenna element **43** such that the antenna element **43** does not contact the body case **10**, and does not necessarily lie on the antenna element **43** with the same shape as the antenna element **43**.

(161) The material forming the dielectric member **63** is not particularly limited as long as it has insulating properties. For example, a resin having a high permittivity may be used. It is preferable that the material, shape and so forth of the dielectric member **63** be set according to the function required for an antenna constituted by the antenna element **43**. The dielectric member **63** disposed as described above can contribute to appropriate wavelength shortening or the like, and cause the antenna element **43** to function as an antenna (e.g. patch antenna).

(162) Preferably, the dielectric member **63** functions as a waterproof packing (sealing member) between the body case **10** and the antenna element **43** and makes the inside of the body case **10** watertight.

(163) Thus, without a separate member, the inside of the body case **10** can be made watertight by the member for preventing the antenna element **43** and the body case **1** as a conductive part from short-circuiting.

(164) When the dielectric member **63** is not required to have a function as a packing that keeps the water-tightness of the body case **10**, the dielectric member **63** may not be disposed over the whole circumference of the upper surface of the body case **10**.

(165) As in the first embodiment, in the body case **10**, the substrate **7** provided with the electric circuit **71** is housed. The antenna element **43** is connected to the electric circuit **71** on the substrate **7** by the first connector **73**.

(166) The first connector **73** is for power supply, and the antenna element **43** is connected, through the first connector **73**, to an LC circuit or the like as the electric circuit **71** mounted on the substrate **7**.

(167) The substrate **7** is disposed on the back cover **3** through a spacer(s) **76** or the like.

(168) When, like this embodiment, the antenna element **43** is disposed, through the dielectric member **63**, over the upper portion of the body case **10** formed of a metal material, thereby configuring an antenna, the characteristics (gain and frequency) of the antenna can be adjusted by

changing various conditions/factors.

(169) Hereinafter, with reference to FIG. 7 to FIG. 11, how to adjust the antenna characteristics, such as the frequency and the gain, when the body case **10** is formed of a metal material will be described.

(170) Graphs shown in FIG. 9 and FIG. 11 show simulation results, which are different from actual measurement values.

(171) In FIG. 7, on the upper surface of the body case **10**, the dielectric member **63** and the antenna element **43** are disposed in this order along the edge of the opening of the body case **10**.

(172) In the example shown in FIG. 7, the antenna element **43** (an antenna element **43a** and a dielectric member **63a** lying thereon) has a width dimension in the radial direction that is approximately half the width dimension of the upper surface of the body case **10** in the radial direction.

(173) In the example shown in FIG. 7, the whole antenna element **43a** is placed over the upper portion of the body case **10**. On the upper portion of the body case **10**, the area of a portion where the antenna element **43a** lies and the area of a portion where the antenna element **43a** does not lie are approximately equal.

(174) In an example shown in FIG. 8A, the inside diameter of the body case **10** is larger than that shown in FIG. 7. As a result of the increase of the inside diameter of the body case **1**, the antenna element **43** (an antenna element **43b** and a dielectric member **63b** lying thereon) does not contact the upper surface of the body case **10** on the inner peripheral side for about half the width of the antenna element **43b**, thereby being in a floating state.

(175) In an example shown in FIG. 8B, the inside diameter of the body case **10** is larger than that shown in FIG. 8A. As a result of the increase of the inside diameter of the body case **1**, the antenna element **43** (an antenna element **43c** and a dielectric member **63c** lying thereon) does not contact the upper surface of the body case **10** almost at all, thereby being in a floating state.

(176) FIG. 9 shows results of a simulation in which the inside diameter of the body case **10** is changed such that the thickness of the body case **10** in the circumferential direction decreases, thereby showing tendencies of change in antenna characteristics.

(177) FIG. 9 shows the frequency (GHz) on the vertical axis on the left side of the graph, the gain (dBc) on the vertical axis on the right side thereof, and the inside diameter (mm) of the body case on the horizontal axis.

(178) In the graph, solid lines show the tendency of change in frequency, and broken lines show the tendency of change in gain. In the example shown in FIG. 9, regarding the results of the simulation conducted by using two types of dielectric members **63** having different permittivities A, B, the solid lines show the tendency of change in frequency, and the broken lines show the tendency of change in gain.

(179) As shown in FIG. 9, in the simulation, both when the dielectric member **63** having the permittivity A was used and when the dielectric member **63** having the permittivity B was used, the following tendencies were confirmed: the larger the inside diameter of the body case **10** is, the higher the frequency is, and also the higher the gain is.

(180) This indicates that the characteristics (frequency and gain) of an antenna configured by including the antennal element **43** can be adjusted by changing the area where the body case **10**, the antenna element **43** and the dielectric member **63** lie on top of one another.

(181) More specifically, regarding the width of the body case **10** in the radial direction on the upper portion thereof, it can be said that, on the upper portion of the body case **10**, the smaller the width (area) of a portion where the dielectric member **63** and the antenna element **43** lie in this order is as compared with the width (area) of a portion where the dielectric member **63** and the antenna element **43** do not lie, the higher the frequency of an antenna configured by including the antenna element **43** is, and also the higher the gain thereof is.

(182) As a method for changing the distance between the body case **10** and the antenna element **43**

to change the antenna characteristics, other than changing the inside diameter of the body case **10**, the following may be used, for example: forming an inner groove in the inner surface of the body case **10**, thereby making the distance between the body case **10** and the antenna element **43** disposed over the upper portion of the body case **10** longer. It is inferred that distancing the inner surface of the body case **10** from the antenna element **43** by, for example, forming an inner groove can exhibit the same effects as those exhibited by making the inside diameter of the body case **10** larger.

(183) Although not shown, when the distance between the body case **10** and the antenna element **43** was changed, the following tendencies were confirmed: the smaller the inside diameter of the antenna element **43** as a bezel member is, the higher the gain is, whereas the larger the inside diameter of the antenna element **43** is, the closer the antenna element **43** is to the body case **10**, and accordingly the lower the gain is.

(184) When the windshield member **2** was fastened by the antenna element **43** as a bezel member, the following tendencies were confirmed: the larger the diameter of the windshield member **2** is, the larger the inside diameter of the antenna element **43** is for the increased amount, and accordingly the lower the gain as an antenna is. When the inside diameter of the antennal element **43** was made to correspond to the diameter of the windshield member **2** by making the inner diameter of the antenna element **43** larger without changing the outside diameter of the antenna element **43**, the width of the antenna element **43** in the radial direction became narrower. Hence, in this case, the following tendencies were confirmed: the gain decreases, but the frequency increases.

(185) Although not shown, when the outside diameter of the body case **10** was changed, both the frequency and the gain of the antenna did not show significant change.

(186) When the height of the body case **10** (distance from the substrate **7** to the back cover **3**) was changed, the frequency did not show significant change, but the gain (sensitivity) showed a tendency to be higher as the height of the body case **10** is higher.

(187) When the outside diameter of the antenna element **43** was changed, the size of the antenna element **43** was larger as the outside diameter was larger, and the frequency showed a tendency to be lower as the outside diameter of the antenna element **43** is larger. The gain did not show significant change.

(188) Further, a simulation was conducted in which GND is set to the substrate **7**, and the GND diameter is changed. When, like this second embodiment, the body case **10** was formed of a metal, both the frequency and the gain showed little change.

(189) In this embodiment, the dielectric member (**63**) is disposed between the antenna element **43** and the body case **10** as a conductive part. The characteristics of an antenna configured by including the antenna element **43** may be adjusted by changing the shape of the dielectric member.

(190) Hereinafter, with reference to FIG. **10A** to FIG. **10D** and FIG. **11**, how to adjust the antenna characteristics, such as the frequency and the gain, by changing the shape of the dielectric member will be described.

(191) FIG. **10A** to FIG. **10D** are main part's perspective views showing variations of the shape of the dielectric member. In FIG. **10A** to FIG. **10D**, in order to show the dielectric member to be seen well, the antenna element **43** and a lead line therefor are indicated by two-dot chain lines.

(192) FIG. **10A** shows a ring-shaped dielectric member **64** (**64a**) that is substantially the same as the antenna element **43** in shape. The dielectric member **64a** has no cut or the like, and is in contact with the antenna element **43** over the entire width (e.g. 2.0 mm) of the dielectric member **64a** in the radial direction. The insulation remaining width as a width where the antenna element **43** and the body case **10** as a conductive part are in an insulated state is 2.0 mm.

(193) FIG. **10B** shows a dielectric member **64** (**64b**) that is divided into a plurality of members and disposed so as to be substantially ring-shaped to correspond to the antenna element **43** in shape. The dielectric member **64b** is not in contact with the antenna element **43** at divided portions **640**. Hence, the insulation remaining width as a width where the antenna element **43** and the body case

10 as a conductive part are in an insulated state is 0 mm.

(194) FIG. **10C** shows a ring-shaped dielectric member **64** (**64c**) that is substantially the same as the antenna element **43** in shape and provided with recesses **641** at points. The positions and the number of the recesses **641** are appropriately set. When the recesses **641** are disposed, the dielectric member **64c** is in contact with the antenna element **43** at the recesses **641** over a width shorter than the width (e.g. 2.0 mm) of the dielectric member **64a** in the radial direction. The insulation remaining width as a width where the antenna element **43** and the body case **10** as a conductive part are in an insulated state is about 1.2 mm.

(195) FIG. **11** shows results of a simulation in which the insulation remaining width of the dielectric member **64** is changed, thereby showing tendencies of change in antenna characteristics.

(196) Similarly to the graph shown in FIG. **9**, FIG. **11** shows the frequency (GHz) on the vertical axis on the left side of the graph, the gain (dBi) on the vertical axis on the right side thereof, and the insulation remaining width (mm) on the horizontal axis thereof.

(197) In the graph, solid lines show the tendency of change in frequency, and broken lines show the tendency of change in gain.

(198) As shown in FIG. **11**, in the simulation, the following tendencies were confirmed: the larger the insulation remaining width is, the lower the frequency is. The gain showed slight change due to the insulation remaining width.

(199) When the insulation remaining width of the dielectric member **64** was changed, both the frequency and the gain did not change dramatically, but showed some change.

(200) The shape of the dielectric member **64** does not appear in the external appearance of the timepiece body. In particular, when, like the insulation remaining width, the shape of the dielectric member **64** in the plane direction is changed, the shape or the like can be changed without the design being affected, relatively easily. Hence, this can be regarded as an effective method for adjusting the antenna characteristics, such as the frequency and the gain, to desired ones.

(201) The method for changing the shape of the dielectric member **64** is not limited thereto.

(202) For example, although in FIG. **10C**, the recesses **641** are formed in the inner peripheral surface of the dielectric member **64**, recesses may be formed in the outer peripheral surface of the dielectric member **64**, or in both the inner peripheral surface and the outer peripheral surface of the dielectric member **64**.

(203) Also, the shape of the recesses may be any. For example, the recesses may be rectangular notches, semicircular notches or the like. The recesses may have different shapes from one another.

(204) As shown in FIG. **10D**, in the recesses **641** shown in FIG. **10C**, second dielectric members **642** having a permittivity different from that of the body **643** of the dielectric member **64** may be fitted. The method for fitting the second dielectric members **642** is not particularly limited. The second dielectric members **642** may not be fitted in all the recesses **641**, and also may not all be the same type.

(205) When the second dielectric members **642** are fitted in the recesses **641** as separate/other members, the antenna characteristics can be finely adjusted to desired frequency and/or gain by appropriately changing the number, size and/or type (permittivity) of the second dielectric members **642**.

(206) Thus, the dielectric member **64** (**64d**) as a whole does not have a cut or unevenness, but is different in permittivity from the dielectric member entirely formed of one type of dielectric. This is inferred to affect the frequency and the gain as an antenna. Hence, this can be regarded an effective method for adjusting the frequency and the gain.

(207) In FIG. **10D**, the second dielectric members **642**, which have a different permittivity from the body **643** of the dielectric member **64**, are fitted in the in-plane direction of the dielectric member **64**, but arrangement of a second dielectric member(s) is not limited thereto. For example, the second dielectric member(s) may be disposed in the layers-stacked direction of the dielectric member **64**.

(208) Further, although not shown, a simulation was conducted in which the thickness of the dielectric member **64** in the height direction is changed. As a result, the following tendencies were confirmed: the thicker the dielectric member **64** is, the higher the frequency is, and also the higher the gain is.

(209) If this method is adoptable giving consideration to the design/external appearance or the like of a timepiece, this can be regarded as one of the effective adjustment methods.

(210) As described above, although there are various factors for adjusting the characteristics (frequency and gain) of an antenna configured by including the antenna element **43**, adjustment of the inside diameter of the body case **10** and adjustment of the shape or the like of the dielectric member (in particular, the shape of the dielectric member in the plane direction) relatively largely affect the frequency and the gain of the antenna while little affecting the external appearance/design of the timepiece body. Hence, these can be regarded as effective methods for adjusting the antenna characteristics to, for example, desired frequency and/or gain.

(211) As in the first embodiment, when the body case **1** is formed of a resin, the antenna element (metal bezel) is fastened to the body case **1** by, for example, being press-fitted into the body case **1**, whereas as in this embodiment, when the body case **10** is formed of a metal, the antenna element can be fastened to the body case **10** by various methods as shown in FIG. **12** to FIG. **22**, for example. In FIG. **12** to FIG. **22**, the same members as those in this second embodiment shown in FIG. **7** or the like are denoted by the same reference signs, and descriptions thereof are omitted. The methods described hereinafter are merely examples, and the method for fastening the antenna element (metal bezel) is not limited thereto.

(212) FIG. **12** is a side sectional view showing an example in which an antenna element is fastened to a body case formed of a metal material. FIG. **13** is an enlarged view of a region XIII enclosed by a chain line in FIG. **12** and shows an example of an antenna element fastening structure.

(213) As shown in FIG. **12** and FIG. **13**, when an antenna element **45** (metal bezel) is fastened to the metal body case **10**, the antenna element **45** is fitted into the opening of the body case **10** through a dielectric member **65**.

(214) In the example shown in FIG. **13**, the antenna element **45** includes a cylindrical part **450**, which is cylindrically shaped, a flange **451** projecting outward from the upper portion of the cylindrical part **450**, and a blind **452** hanging from the outer peripheral end surface of the flange **451**. In the cylindrical part **450**, the windshield member **2** is fitted.

(215) The dielectric member **65** includes: a dielectric body **651** disposed between the lower surface of the flange **451** of the antenna element **45** and the upper surface of the body case **10**; and a cylindrical part **652** projecting downward from the inner end surface of the dielectric body **651** along the inner peripheral surface and disposed between the outer peripheral surface of the cylindrical part **450** of the antenna element **45** and the upper portion of the inner peripheral surface of the body case **10**.

(216) In this example, the dielectric member **65** is capable of fastening the antenna element **45** by deforming with some elasticity and generating stress, and is formable with various resins.

(217) The antenna element **45** is press-fitted into the body case **10** through the dielectric member **65**, and the dielectric member **65** functions as a waterproof packing (sealing member) that makes the inside of the body case **10** watertight.

(218) Thus, when the dielectric member **65** doubles as a waterproof packing (sealing member), a separate waterproof packing (sealing member), such as an O-ring, for the water-tightness is unneeded. This can reduce the number of parts.

(219) FIG. **14** is a side sectional view showing a modification of the antenna element fastening structure shown in FIG. **12** and FIG. **13**. In this modification, the configuration of the dielectric member is different from that shown in FIG. **12** and FIG. **13**.

(220) In FIG. **14**, a dielectric member **66** is formed of a hard resin that hardly deforms, and includes: a dielectric body **661** disposed between the lower surface of the flange **451** of the antenna

element **45** and the upper surface of the body case **10**; and a cylindrical part **662** projecting downward from the inner end surface of the dielectric body **661** along the inner peripheral surface and disposed between the outer peripheral surface of the cylindrical part **450** of the antenna element **45** and the upper portion of the inner peripheral surface of the body case **10**.

(221) On the inner peripheral surface and the outer peripheral surface of the cylindrical part **662**, threaded portions **663** are formed by thread-cutting.

(222) On the outer peripheral surface of the cylindrical part **450** of the antenna element **45**, a threaded portion **453** that meshes with the threaded portion **663** formed on the inner peripheral surface of the cylindrical part **662** is formed, so that the threaded portion **453** of the antenna element **45** and the threaded portion **663** of the dielectric member **66** mesh with one another.

(223) On the upper portion of the inner peripheral surface of the body case **10**, a threaded portion **101** that meshes with the threaded portion **663** formed on the outer peripheral surface of the cylindrical part **662** is formed, so that the threaded portion **101** of the body case **10** and the threaded portion **663** of the dielectric member **66** mesh with one another.

(224) The dielectric member **66** is screwed on the body case **10** (fastened thereto with the threaded portions (**663**, **101**), which hereinafter may be referred to as “thread-fastened”), and the antenna element **45** is screwed on this dielectric member **66** (fastened thereto with the threaded portions (**453**, **663**)), so that the antenna element **45** is thread-fastened to the body case **10** through/with the dielectric member **66**.

(225) When the antenna element **45** is thread-fastened as described above, for example, as shown in FIG. **15** or FIG. **16**, an O-ring(s) may be provided as a sealing member.

(226) That is, in FIG. **15**, O-rings **665** are disposed between the upper surface of the dielectric body **661** and the lower surface of the flange **451** of the antenna element **45** and between the lower surface of the dielectric body **661** and the upper surface of the body case **10**. Two O-rings **665** may not be disposed, and hence one O-ring **665** may be disposed on either the upper surface side of the dielectric body **661** or the lower surface side thereof.

(227) In FIG. **16**, the outward projecting width of the dielectric body **661** is made smaller than that of the flange **451** of the antenna element **45**, and outside the outer peripheral surface of the dielectric body **661**, an O-ring **666** is interposed between the lower surface of the flange **451** of the antenna element **45** and the upper surface of the body case **10**.

(228) When an O-ring(s) are provided as a sealing member(s) as shown in FIG. **15** or FIG. **16**, the body case **10** can be made watertight with higher certainty.

(229) The dielectric member is not limited to a single member.

(230) For example, as shown in FIG. **17**, there may be provided, as separate members, a dielectric body **671** disposed between the lower surface of the flange **451** of the antenna element **45** and the upper surface of the body case **10**, and a cylindrical part **672** that is disposed between (i) the inner end surface of the dielectric body **671** and the upper portion of the inner peripheral surface of the body case **10** and (ii) the outer peripheral surface of the cylindrical part **450** of the antenna element **45**, and has a threaded portion **673** on the inner peripheral surface.

(231) The threaded portion **453** formed on the outer peripheral surface of the cylindrical part **450** of the antenna element **45** meshes with and is screwed on the threaded portion **673** formed on the inner peripheral surface of the cylindrical part **672**. The upper portion of the body case **10** is provided with a projection **105** projecting inward, and the lower end portion of the cylindrical part **672** is provided with a flange **675** projecting outward. The flange **675** is fastened to the lower surface of the projection **105** of the body case **10**, so that the antenna element **45** is thread-fastened to the body case **10** through the dielectric member **67** (the dielectric body **671** and the cylindrical part **672**).

(232) Further, for example, as shown in FIG. **18**, there may be provided, as separate members, a dielectric body **681** disposed between the lower surface of the flange **451** of the antenna element **45** and the upper surface of the body case **10**, and a cylindrical part **682** that is disposed (i) on the

inner peripheral side of the lower surface of the dielectric body **681** and (ii) between the upper portion of the inner peripheral surface of the body case **10** and the outer peripheral surface of the cylindrical part **450** of the antenna element **45**, and has a threaded portion **683** on the inner peripheral surface.

(233) The threaded portion **453** formed on the outer peripheral surface of the cylindrical part **450** of the antenna element **45** meshes with and is screwed on the threaded portion **683** formed on the inner peripheral surface of the cylindrical part **682**. The upper portion of the body case **10** is provided with the projection **105** projecting inward, and the lower end portion of the cylindrical part **682** is provided with a flange **685** projecting outward. The flange **685** is fastened to the lower surface of the projection **105** of the body case **10**, so that the antenna element **45** is thread-fastened to the body case **10** through the dielectric member **68** (the dielectric body **681** and the cylindrical part **682**).

(234) Thus, by dividing the dielectric member into a plurality of members, the members of the dielectric member can be formed of different materials from one another.

(235) In particular, when, like the example shown in FIG. **18**, there are separate members, namely, the dielectric body **681** and the member (cylindrical part **682**) having the threaded portion **683**, for example, the cylindrical part **682** having the threaded portion **683** may be formed of a hard material, and the dielectric body **681** may be formed of an elastic deformable material, so that the antenna element **45** can be thread-fastened to the body case **10** through the cylindrical part **682**, and also the water-tightness can be ensured by making the dielectric body **681** function as a sealing member (waterproof packing).

(236) When a structure, as shown in FIG. **17** or FIG. **18**, in which a positional relationship between a body case and an antenna element (bezel) can be fixed is adopted, the body case, the dielectric member and the antenna element may be provided with alignment parts that are fitted with one another. In this manner, when the cylindrical part (**672** in FIG. **17** or **682** in FIG. **18**) of the dielectric member is inserted (twisted) from the back side (back cover side) of a timepiece, a mark engraved at the 12 o'clock position on the antenna element (bezel) can be matched with the 12 o'clock position on the body case, for example. Because a module as an inner structure having a dial plate and/or the like of the timepiece is also aligned with the body case, the 12 o'clock position on the dial plate can also be matched with the mark at the 12 o'clock position on the antenna element (bezel). Thus, their alignment is easily completed.

(237) Further, as shown in FIG. **19** to FIG. **22**, an antenna element **46**, a dielectric member **69** and the body case **10** may be fastened with a screw(s).

(238) That is, for example, as shown in FIG. **19**, the antenna element **46** is provided with a through hole **461**, and the dielectric member **69** is provided with a hole **691** at a position corresponding to the through hole **461**. A screw **91** is inserted from above the antenna element **46** into the hole **691** through the through hole **461**, so that the antenna element **46** and the dielectric member **69** are fastened with the screw **91**.

(239) A recess may be provided around the through hole **461** of the antenna element **46** so that the head of the screw **91** can be disposed in the recess. In this case, the head of the screw **91** does not project from the surface of the antenna element **46**, so that a beautiful external appearance can be obtained.

(240) The dielectric member **69** and the body case **10** are fastened with a screw **92**. More specifically, the dielectric member **69** is provided with a through hole **693**, and the upper portion of the through hole **693** is formed to be a large-diameter part **692**, the diameter of which is larger than that of the head of the screw **92**. The body case **10** is provided with a hole **102** at a position corresponding to the through hole **693**, which includes the large-diameter part **692**. The screw **92** is inserted from above the dielectric member **69** into the hole **102** through the through hole **693** (**692**), so that the dielectric member **69** and the body case **10** are screw-fastened. Although the depth of the through hole **693** (**692**) formed in the dielectric member **69** is not particularly limited, it is

preferable that the through hole **693** be formed deep enough for the screw **92** not to float and contact the antenna element **46**. In the through hole **693 (692)**, a member formed of an insulator (dielectric) may be interposed between the outer peripheral surface of the screw **92** and the through hole **693**, or after the screw **92** is inserted, an adhesive (sealant) or the like formed of an insulating (dielectric) material may be put in the through hole **693 (692)**, or a member formed of an insulator (dielectric) may be fitted therein. This can prevent the screw **92** from contacting the antenna element **46** for sure.

(241) To fasten the antenna element **46**, the dielectric member **69** and the body case **10** with the screws **91, 92**, first, the dielectric member **69** and the body case **10** are screw-fastened, next, the antenna element **46** is disposed thereon, and then the dielectric member **69** and the antenna element **46** are screw-fastened.

(242) Further, for example, as shown in FIG. **20**, an exterior member **70** covering the antenna element **46** may also be provided. The exterior member **70** is a decorative member formable with various materials, such as resin and ceramic.

(243) The exterior member **70** is provided with a through hole **701**. It is preferable that a recess **702** be provided around the through hole **701** so that the head of the screw **91** can be disposed in the recess **702**. Because the exterior member **70** is the outermost exterior member of the timepiece body, a beautiful external appearance can be obtained by preventing the head of the screw **91** from projecting from the exterior member **70**.

(244) The antenna element **46** is provided with a through hole **462** at a position corresponding to the through hole **701**. The through hole **462** is formed to have a diameter large enough for the screw **91** not to contact the antenna element **46** in the through hole **462**.

(245) The dielectric member **69** is provided with a hole **691** at a position corresponding to the through hole **701** and the through hole **462**.

(246) The screw **91** is inserted from above the exterior member **70** into the hole **691** through the through hole **701** and the through hole **462**, so that the exterior member **70**, the antenna element **46** and the dielectric member **69** are screw-fastened.

(247) In the through hole **462** of the antenna element **46**, a member formed of an insulator (dielectric) may be interposed between the outer peripheral surface of the screw **91** and the through hole **462**, or after the screw **91** is inserted, an adhesive (sealant) or the like formed of an insulating (dielectric) material may be put in the through hole **462**. This can prevent the screw **91** from contacting the antenna element **46** for sure.

(248) The dielectric member **69** and the body case **10** are fastened with the screw **92** in the same manner as that described with reference to FIG. **19**.

(249) Further, for example, as shown in FIG. **21**, when the exterior member **70** covering the antenna element **46** is provided, members from the exterior member **70** to the body case **10** may be fastened with one (long) screw **93**.

(250) The exterior member **70** is provided with a through hole **703**. It is preferable that a recess **704** be provided around the through hole **703** so that the head of the screw **93** can be disposed in the recess **704**.

(251) The antenna element **46** is provided with a through hole **463** at a position corresponding to the through hole **703**. The through hole **463** is formed to have a diameter large enough for the screw **93** not to contact the antenna element **46** in the through hole **463**.

(252) The dielectric member **69** is provided with a through hole **694** at a position corresponding the through hole **703** and the through hole **463**.

(253) The body case **10** is provided with a hole **103** at a position corresponding to the through hole **703**, the through hole **463** and the through hole **694**.

(254) The screw **93** is inserted from above the exterior member **70** into the hole **103** through the through hole **703**, the through hole **463** and the through hole **694**, so that the exterior member **70**, the antenna element **46**, the dielectric member **69** and the body case **10** are screw-fastened.

(255) In the through hole **463** of the antenna element **46**, a member formed of an insulator (dielectric) may be interposed between the outer peripheral surface of the screw **93** and the through hole **463**, or after the screw **93** is inserted, an adhesive (sealant) or the like formed of an insulating (dielectric) material may be put in the through hole **463**. This can prevent the screw **93** from contacting the antenna element **46** for sure.

(256) When no exterior member is provided, in order that the screw **93** be not in direct contact with the antenna element **46**, an insulating coating or the like may be disposed on the head of the screw **93** and/or a portion of the screw **93** disposed in the antenna element **46**, or as shown in FIG. **22**, an insulating washer **705** or the like formed of an insulating material may be disposed in a through hole **464** of the antenna element **46**. This can prevent the screw **93** and the antenna element **46** from short-circuiting.

(257) As described above, according to this embodiment, the timepiece body (**100**) as an electronic device includes the ring-shaped antenna element **43** and the body case **10** that is formed of a metal material and over the upper portion of which the antenna element **43** is disposed, and the dielectric member **63** is interposed between the antenna element **43** and the body case **10**, and the body case **10** functions as a conductive part having the same potential as GND.

(258) Thus, because the body case **10** itself is a conductive part, a separate conductive part is unneeded.

(259) Further, when the body case **10** is formed of a metal, the timepiece body **100** having a high-quality external appearance and an excellent design and a wrist device (e.g. wristwatch) including this timepiece body **100** can be produced.

(260) Further, because the dielectric member **63** is interposed between the antenna element **43** and the body case **10**, the characteristics, such as the frequency and the gain, of an antenna configured by including the antenna element **43** can be easily adjusted to desired characteristics by this dielectric member **63** or the like.

(261) Further, the dielectric member **63** may double as a sealing member that makes the inside of the body case **10** watertight.

(262) In this case, a watertight structure can be produced without a separate O-ring or the like.

(263) Further, the characteristics of an antenna configured by including the antenna element **43** may be adjusted by changing the area where the body case **10** and the antenna element **43** lie on (over) top of one another.

(264) For example, the body case **10** is formed such that on the upper portion of the body case **10**, the area of a portion where the antenna element **43** lies is smaller than the area of a portion where the antenna element **43** does not lie.

(265) In this case, the antenna characteristics can be adjusted by a method little affecting the external appearance. More specifically, the gain of the antenna can be made higher. This makes designability of an electronic device, such as a timepiece, and adjustment of the frequency and/or the gain of an antenna compatible.

(266) Further, the characteristics of an antenna configured by including the antenna element **43** may be adjusted by changing the distance between the body case **10** and the antenna element **43**.

(267) In this case too, the antenna characteristics can be adjusted by a method little affecting the external appearance. This makes designability of an electronic device, such as a timepiece, and adjustment of the frequency and/or the gain of an antenna compatible.

(268) Further, the characteristics of an antenna configured by including the antenna element **43** may be adjusted by changing the shape of the dielectric member **63**.

(269) Because the dielectric member **63** is a member that is unlikely to appear in the external appearance, in this case too, the antenna characteristics can be adjusted by a method little effecting the external appearance. This makes designability of an electronic device, such as a timepiece, and adjustment of the frequency and the gain of an antenna compatible.

(270) As a method for changing the shape of the dielectric member **64**, the recesses **641** may be

formed in a peripheral surface(s) of the dielectric member **64**.

(271) Thus, the antenna characteristics can be finely adjusted with a relatively simple method.

(272) When the recesses **641** are formed in a peripheral surface(s) of the dielectric member **64**, the second dielectric members **642** having a permittivity different from the body **643** of the dielectric member **64** may be fitted in the recesses **641**.

(273) Thus, the antenna characteristics can be finely adjusted with a relatively simple method.

(274) The dielectric member **65** may be formed of an elastic deformable material, and the antenna element **45** may be press-fitted into, thereby being fastened to, the body case **10** through the dielectric member **65**.

(275) This can fasten the antenna element **45** to the body case **10** without increasing the number of parts even when the body case **10** is formed of a metal.

(276) Also, the water-tightness can be ensured by the antenna element **45** being press-fitted into, thereby being fastened to, the body case **10** through the dielectric member **66**.

(277) The dielectric member **66** may be provided with the threaded portions **663** on the surface on the contact side with the antenna element **45** and on the surface on the contact side with the body case **10**, and the antenna element **45** may be thread-fastened to the body case **10** through the dielectric member **66**.

(278) Thus, the antenna element **45** and the body case **10** can be fastened by the dielectric member **66** without a separate member.

(279) The antenna element **46**, the dielectric member **69** and the body case **10** may be fastened through/with one or more of the screws **91**, **92**, **93**.

(280) Thus, the antenna element **46**, the dielectric member **69** and the body case **10** can be fastened with a relatively simple method.

(281) Although some embodiments or the like of the present disclosure have been described above, it is needless to say that the present invention is not limited to these embodiments or the like, and various modifications can be made without departing from the scope thereof.

(282) For example, in each of the above embodiments or the like, the timepiece body **100** is described as an electronic device, and the timepiece (wristwatch) is described as a wrist device including the electronic device. However, the electronic device of the present disclosure is not limited to timepieces (wristwatches) but applicable to a wide range of devices including antennas.

(283) For example, the electronic device of the present disclosure may be applied to a variety of devices, such as a pedometer, an altimeter or a barometer, or may be applied to a variety of terminal devices, such as a smartphone.

(284) Although several embodiments or the like of the present disclosure have been described above, the scope of the present invention is not limited to the embodiments or the like described above, but includes the scope of the present invention described in the claims below and the scope of their equivalents.

Claims

1. An electronic device comprising: a ring-shaped antenna element; a conductor having a potential same as GND; and a dielectric interposed between the antenna element and the conductor, wherein the antenna element, the dielectric and the conductor are disposed so as to lie on top of one another in plan view, and wherein the dielectric includes a plurality of dielectrics.
2. The electronic device according to claim 1, comprising a body case disposed below the antenna element and functioning as the conductor having the potential same as the GND.
3. The electronic device according to claim 2, wherein the plurality of dielectrics includes a first dielectric having a recess on a peripheral surface and a second dielectric disposed in the recess and having a permittivity different from the first dielectric.
4. The electronic device according to claim 2, wherein the plurality of dielectrics includes a third

- dielectric disposed between the antenna element and the body case and a fourth dielectric contacting the third dielectric on a first surface and contacting the antenna element on a second surface different from the first surface, and wherein a threaded portion is formed on the second surface of the fourth dielectric to fasten the fourth dielectric to the antenna element.
5. The electronic device according to claim 4, wherein the third dielectric is less harder than the fourth dielectric.
6. The electronic device according to claim 2, wherein the body case is formed of a metal material.
7. A wrist device comprising the electronic device according to claim 1.
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